Malware Analysis Intro

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Outline

Small Introduction

What is malware analysis
- Types of Malware
- How do they work?

Types of Malware Analysis

A simple day...

Consider this scenario:
When you are working or watch a movie online or even click the href in a spam mail, you accidently downloaded a malware in your PC. Now the panic starts, your computer behaves differently.

What do you do?

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Definition

Malware Analysis refers to identify the malware, understand how it works and to how to defeat it eventually.

What is malware?

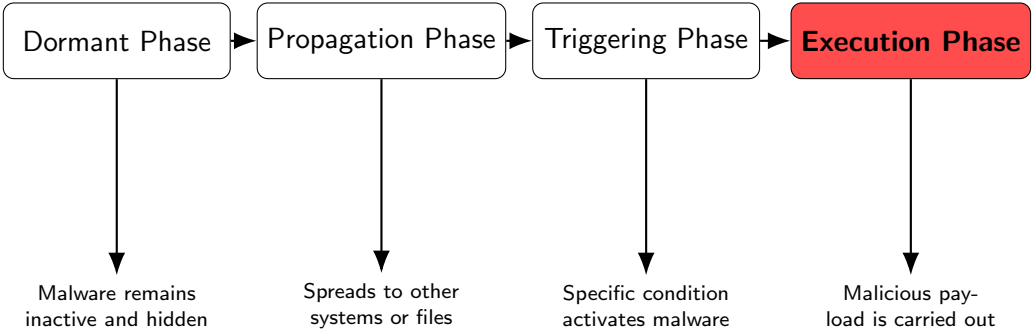
It is a program that can disrupt computer operations, steal sensitive info, gain access to computer systems and spy users.

Key Malware Types

There are many categories of malware out but the most important are:

- Virus: It attaches itself in a program and spreads in other programs
- Trojan: It includes additive unexpected functionality.
- Worm: It spreads itself in other PCs.
- Backdor : It can grand access to an non aythorized user.
- Botnet: A botnet is a network of compromised computers or devices (bots) controlled remotely by a single operator.
- Ransomware:Ransomware is a type of malware that holds a victim's sensitive data or device hostage, threatening to keep it locked
- Spyware: Collects information from a computing system without your consent
- Keyloggers: Is designed to record every key pressed on a keyboard.

Life cycle of malware



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Malware Analysis Types

The are different ways to find and indentify malicious code.

1. **Static Analysis:** It allows us to analyze the malware without executing it. Most of the times the tools we use,help us to find strings,functions or even sumbols inside the malicious code.
2. **Dynamic Analysis:**This method requires to execute the code in order to monitor its activity inside the computer system. We achieve thiw by looking the file system, the process activity, network and system calls.
3. **Memory Analysis:** Memory forensics dives into the volatile memory, aiming to uncover evidence of malware infections and other illicit activities that leave footprints in a system's RAM (Memory Analysis info)